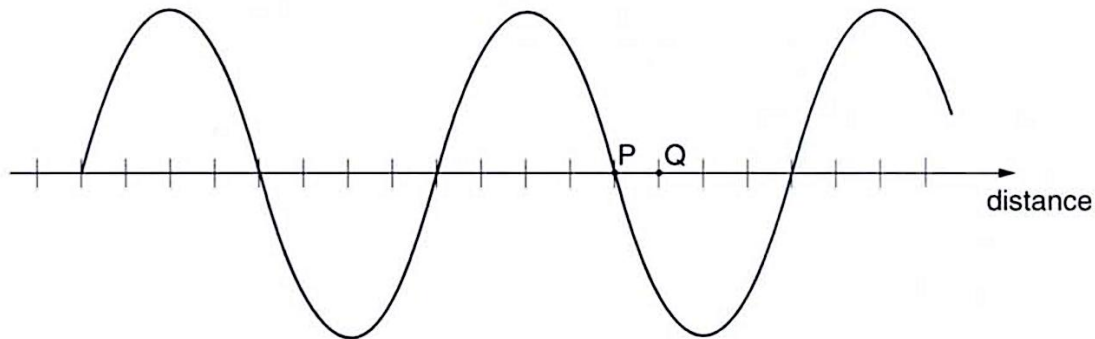


- 17 The diagram shows a transverse wave at a particular instant. The wave is travelling to the right. The frequency of the wave is 25 Hz.



At the instant shown the displacement is zero at the point P.

What is the shortest time to elapse before the displacement is zero at point Q?

A	5 ms	B	15 ms	C	25 ms	D	45 ms
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Answer: A

period = $1/25 = 40$ ms

wavelength is 8 parts

since wave travels to right at constant speed, time taken to traverse 1 part = 5 ms

so, zero at P at the instant shown will be at Q in 5 ms

18 When two polarisers P and Q are placed so that their polarising directions are parallel, the